

Failure Modes and Reliability Boundaries of Cross-Dataset Transfer Learning for Lithium-Ion Battery SOH Estimations

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Abstract—Reliable State-of-Health (SOH) estimation is essential for lithium-ion batteries deployed in electrified transport, grid-integrated storage, and Vehicle-to-Grid/Vehicle-to-Building applications. Although data-driven SOH estimators can achieve high accuracy within individual datasets, their reliability under cross-dataset deployment is limited by shifts in chemistry, operating protocol, feature availability, and prediction horizon. This paper investigates the failure modes and reliability boundaries of cross-dataset transfer learning for Long Short-Term Memory-based SOH estimation. Five public BatteryLife datasets are evaluated through 20 source-target combinations using zero-shot transfer, head-only fine-tuning, and full fine-tuning at multiple target-data fractions, resulting in 140 main transfer experiments. Results show that zero-shot transfer fails in all evaluated pairs, with RMSE degradation ranging from 51% to 3242%, indicating that direct deployment without target-domain adaptation is unreliable. Fine-tuning improves performance under identifiable conditions: the data-scarce CALCE target obtains approximately 20% RMSE improvement from all non-CALCE sources, whereas MATR consistently resists transfer. The evaluated targets further exhibit an empirical two-regime pattern, with some targets dominated by intrinsic target properties and others sensitive to source selection. A finer-grained fraction sweep reveals a threshold-like activation pattern, and prediction-horizon mismatch appears as a recurring cross-dataset failure mode. These findings provide reliability-oriented guidance for source selection, target-data collection, and safe deployment of transfer learning in battery SOH monitoring.

Index Terms—Lithium-Ion Batteries, State of Health, Transfer Learning, Cross-Dataset Generalisation, System Reliability, Battery Safety

I. INTRODUCTION

Lithium-ion batteries (LIBs) are increasingly deployed in electric vehicles (EVs), grid-scale storage, and building energy systems. As Vehicle-to-Grid (V2G) and Vehicle-to-Building (V2B) technologies enable EV batteries to supply energy back

to grids and buildings [1], inaccurate degradation assessment may lead to unsafe discharge depth, unexpected capacity loss, or reduced power availability [2]. Accurate real-time State-of-Health (SOH) monitoring is therefore essential for battery safety and system reliability.

In energy storage applications, machine learning has emerged as a promising approach for estimating battery state of charge, state of health, and remaining useful life [3]. However, data-driven models require substantial training data to achieve reliable performance, even though they avoid explicit electrochemical modelling [4]. Their reliability often degrades under cross-dataset deployment, as battery ageing trajectories vary across chemistries, manufacturers, cycling protocols, operating conditions, and available feature sets [5]. This dataset-specificity remains a critical barrier to safe and scalable BMS deployment.

Transfer learning offers a principled route for improving cross-dataset SOH estimation by transferring knowledge from a data-rich source domain to a data-limited target domain, thereby reducing the need to train a target-domain model from scratch [6]. Despite its potential, the reliability of transfer learning under broad cross-dataset deployment remains insufficiently characterised, particularly when source and target datasets differ in chemistry, feature availability, dataset scale, and prediction horizon [7]. These limitations make it difficult to derive practical guidance for deploying SOH models across heterogeneous battery datasets.

This paper investigates the failure modes and reliability boundaries of cross-dataset transfer learning for LSTM-based LIB SOH estimation. The main contributions are summarised as follows:

- 1) A reliability-oriented all-pairs evaluation across five

Unified Framework for Cross-Dataset SOH Prediction via Transfer Learning

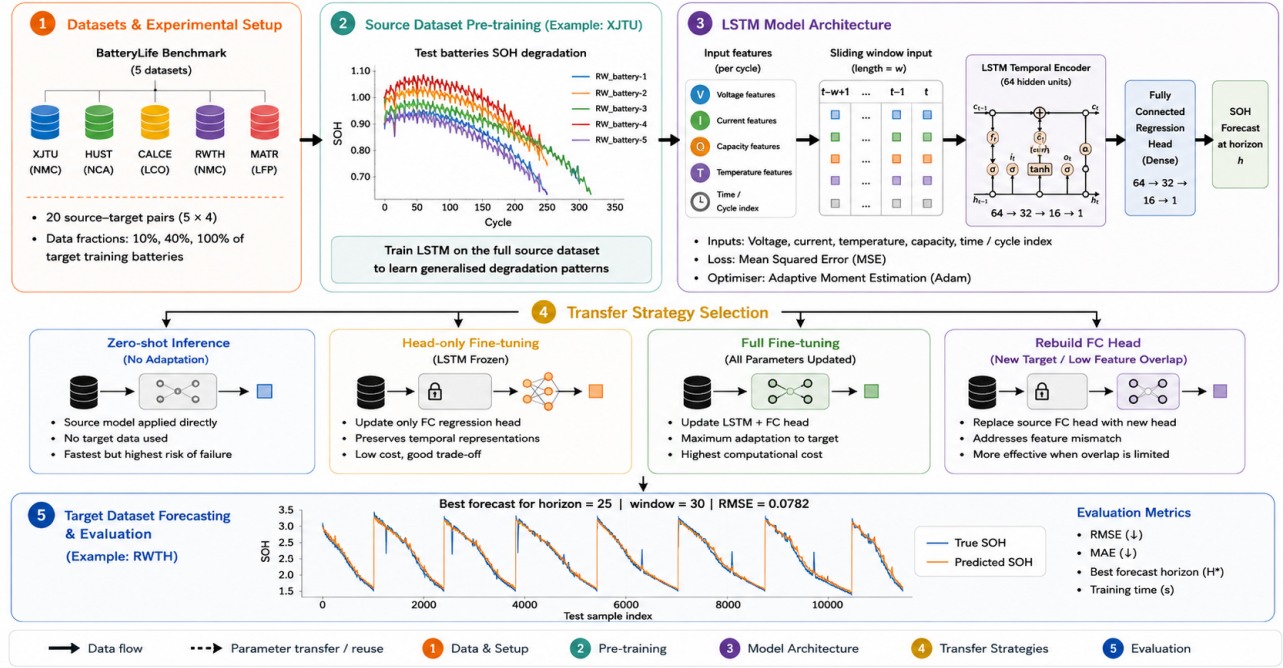


Fig. 1. Overview of the proposed cross-dataset SOH prediction framework. The workflow includes: (1) BatteryLife dataset setup across five datasets; (2) source-dataset pre-training using degradation trajectories; (3) LSTM model architecture with sliding-window inputs and an FC regression head; (4) transfer strategy selection, including zero-shot inference, head-only fine-tuning, full fine-tuning, and FC-head rebuilding; and (5) target-domain SOH forecasting and evaluation.

BatteryLife-derived datasets, covering 20 source-target pairs and 140 transfer experiments with zero-shot, head-only fine-tuning, and full fine-tuning strategies.

- 2) An empirical characterisation of transfer reliability boundaries, including source-dependent and source-independent target behaviours and a threshold-like activation pattern under limited target data.
- 3) An analysis of key cross-dataset failure modes, including prediction-horizon mismatch and feature-content mismatch, with guidance for source selection and safer SOH model deployment.

II. RELATED WORK

A. Battery Degradation and Reliability

Battery SOH reduction arises from chemistry-specific electrochemical ageing mechanisms [8]. Solid Electrolyte Interface (SEI) growth at the graphite anode consumes lithium inventory; lithium plating under high-rate charging creates dendrite-related safety risks. At the cathode, NMC cells suffer particle cracking; LFP cells exhibit lithium ordering phenomena; LCO cells are susceptible to structural collapse [1]. These mechanisms and variations in operating profiles produce fundamentally different degradation trajectories across datasets, making cross-dataset reliability modelling inherently challenging, particularly under stochastic battery operation [9]. Data-driven methods learn degradation patterns directly from cycling data [10], but require large, dataset-specific training sets.

B. LSTM-Based SOH Estimation

Zhang et al. [11] demonstrated LSTM for battery remaining useful life prediction, showing the gating mechanism retains long-range temporal dependencies across hundreds of cycles. Wang et al. [12] conducted a comparative study of battery SOH prediction models, exploring Transformer-based approaches with reversible instance normalisation as an alternative to LSTM for sequence-based SOH estimation. Guo et al. [13] further combined MBLSTM and iTransformer models with anomaly detection and explainability for SOH prediction under heterogeneous operating conditions. Huang et al. [14] proposed a combined health indicator framework with LSTM, achieving RMSE below 1.44% across multiple datasets, and introduced fine-tuning and rebuilding transfer strategies that directly motivate this work. Lu et al. [15] demonstrated SOH estimation from partial charging curves without additional degradation experiments. Li et al. [16] showed that feature fusion with a hybrid CNN-LSTM architecture improves robustness under varying operating conditions.

C. Transfer Learning for Battery SOH

Ren et al. [5] demonstrated LSTM-based transfer learning reduces target-domain data requirements, characterising transfer benefit as a continuous function of data fraction. Qi et al. [7] extended transfer to EIS-based features for laboratory-to-field adaptation. Yao et al. [17] explored semi-supervised adversarial transfer for battery capacity estimation. The key gap across these studies is the absence of systematic all-pairs

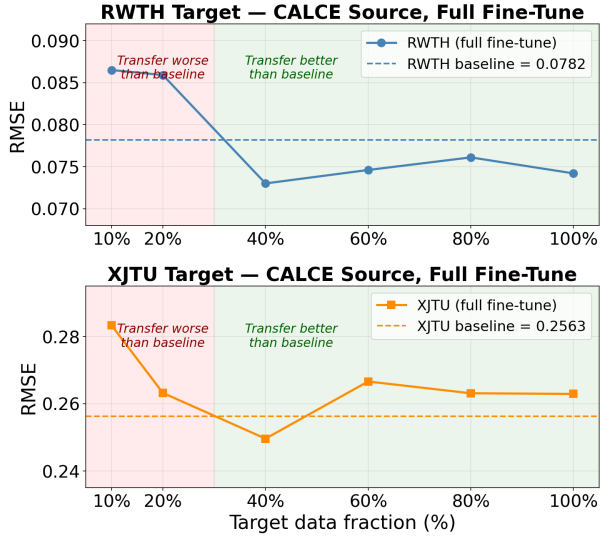


Fig. 2. Transfer activation threshold — CALCE source, full fine-tuning. RMSE compared to target data fraction for RWTH (top) and XJTU (bottom). Dashed lines show from-scratch baselines. Both targets cross below baseline at 40%; for XJTU, RMSE returns above baseline beyond 40% due to feature-space mismatch between CALCE and XJTU degradation dynamics.

evaluation across diverse chemistries at scale — preventing characterisation of whether transfer effectiveness is driven by source or target dataset properties, and whether failure modes such as horizon mismatch are dataset-specific or general.

III. METHODOLOGY

A. SOH Definition and Datasets

SOH is defined as the ratio of current to rated discharge capacity:

$$\text{SOH}(k) = \frac{Q_k}{Q_{\text{rated}}} \times 100\% \quad (1)$$

where Q_k is the discharge capacity at cycle k . A battery is considered end-of-life when SOH falls below 80% [8]. In V2G/V2B contexts, accurate SOH estimation near this threshold is safety-critical — an overestimate risks unsafe discharging of a degraded cell into a building load.

Five publicly available datasets are used, selected to maximise diversity in chemistry, feature richness, and operating conditions. Table I summarises their properties. CALCE is severely data-scarce (six LCO batteries, three features). MATR has the richest feature set (18 signals including DC internal resistance, energy efficiency, and charge throughput) encoding LFP fast-charge degradation dynamics absent from all other datasets. RWTH is the largest by battery count (668 cells) but reports capacity in Ah rather than normalised SOH. Datasets are sourced from the BatteryLife benchmark [18], which provides a standardised preprocessing and evaluation format for cross-dataset battery research.

B. LSTM Architecture and Training

The LSTM model consists of a single LSTM layer (hidden size 64), a dropout layer (rate 0.20), and three fully connected

TABLE I
SUMMARY OF DATASETS

Dataset	Cells	Features	Max cycles	Chemistry
XJTU	55	15	~350	NMC
HUST	77	13	~2000	LFP
CALCE	6	3	~1100	LCO
RWTH	668	15	~160	Mixed
MATR	139	18	~2000	LFP

(FC) layers [64, 32, 16] producing a scalar SOH output — 27,521 trainable parameters, as illustrated in Panel 3 of Fig. 1. The cell state and hidden state are updated via:

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c) \quad (2)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (3)$$

where $\mathbf{f}_t$, $\mathbf{i}_t$, $\mathbf{o}_t$ are the forget, input, and output gate activations. Input sequences use a sliding window: $\hat{\mathbf{y}}_{t+h} = f_{\theta}(\mathbf{x}_{t-w+1}, \dots, \mathbf{x}_t)$ where w is the window size and h is the forecast horizon. Training uses Adam (lr=0.001), MSE loss, and early stopping (patience=8). Each dataset is split 70/15/15 by battery ID — preventing leakage and ensuring reliability metrics reflect genuine generalisation to unseen batteries.

C. Transfer Learning Strategies

All 20 source–target pair combinations are evaluated under three transfer settings, as illustrated in Panel 4 of Fig. 1. Zero-shot transfer applies the pretrained source model directly to the target test set without adaptation, representing the fastest but most failure-prone deployment case. Head-only fine-tuning freezes the LSTM backbone and retrain only the fully connected regression layers, preserving source-domain temporal representations while adapting the output mapping. Full fine-tuning updates all parameters on target data, providing the greatest adaptation capacity but also the highest risk of overfitting or catastrophic forgetting in low-data target regimes.

Both fine-tuning strategies are evaluated at 10%, 40%, and 100% of target training batteries in the main 20-pair sweep (7 configurations per pair, 140 total). A finer-grained fraction sweep at 10%, 20%, 40%, 60%, 80%, and 100% is additionally conducted using CALCE as source to precisely characterise the transfer activation threshold. Feature alignment uses the intersection of shared features between source and target. When fewer than five features are shared, the LSTM backbone is randomly reinitialised and only FC head weights are seeded from the source model.

IV. RESULTS

A. Baseline Performance

A systematic baseline sweep is conducted over window size $w \in \{10, 20, 30, 40, 50\}$ cycles and forecast horizon $h \in \{1, 10, 25, 50\}$ cycles, producing 100 configurations per dataset. The window size w controls how many historical cycles the LSTM receives as input, while the forecast horizon h determines how many cycles ahead the model predicts SOH.

All 20 window-horizon combinations are evaluated on the held-out test split and the optimal configuration per dataset — that achieving minimum test RMSE — is fixed as the source pre-training configuration for all subsequent transfer learning experiments involving that dataset. Table II reports the best RMSE per dataset. Fig. 3 shows RMSE heatmaps for RWTH and MATR, confirming that no single configuration generalises across datasets. HUST and MATR achieve the lowest RMSE (0.0095 and 0.0085), reflecting large clean training sets. XJTU achieves the highest RMSE (0.2563) from high cycle-to-cycle variability. A computational cost study confirms performance saturates before the full training set: for HUST, 40% data achieves RMSE=0.0100 at 447 s versus RMSE=0.0126 at 608 s for full training — a 42% reduction in training time with improved accuracy.

TABLE II
BEST BASELINE LSTM RESULTS PER DATASET

Dataset	W	H	RMSE	MAE
XJTU	30	1	0.2563	0.1802
HUST	30	10	0.0095	0.0073
CALCE	20	25	0.1998	0.0718
RWTH	30	25	0.0782	0.0602
MATR	50	10	0.0085	0.0065

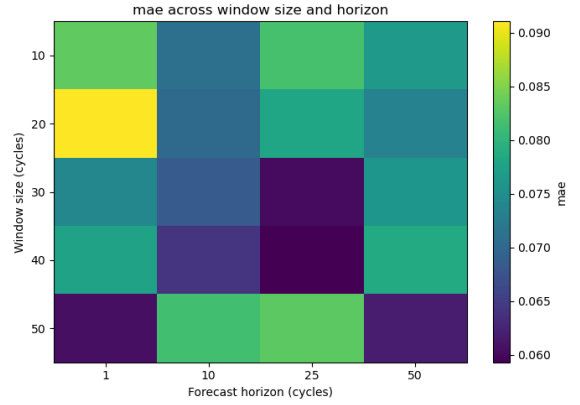
B. Transfer Learning Results

Table III summarises the best RMSE improvement over the from-scratch baseline for all 20 source-target pairs. Zero-shot transfer fails universally — RMSE degrades by 51%–3242% across all 20 pairs, with zero-shot R^2 reaching as low as -11.99 on RWTH, confirming predictions are actively worse than a simple mean predictor.

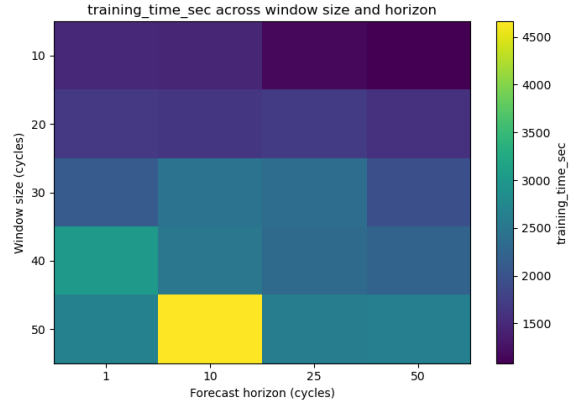
Fine-tuning consistently recovers and frequently exceeds baseline performance. CALCE achieves +20% improvement from all five source datasets, ranging from +20.6% (XJTU) to +21.7% (RWTH) — a narrow band across sources spanning NMC, NCA, LCO, LFP, and mixed chemistries. Fig. 4 shows per-source heatmaps for HUST and MATR as sources, confirming the CALCE improvement and MATR resistance patterns hold regardless of source. Table IV provides detailed RMSE results for CALCE as target. The RWTH full fine-tune at 10% diverges to RMSE=0.6974 due to prediction horizon mismatch (source H=25, target H=10) combined with very limited target data — a result that foreshadows Finding 4.

C. Transfer Benefit Activation and Data Threshold Analysis

A finer-grained fraction sweep with CALCE as source (Table V) reveals that for both RWTH and XJTU as targets, RMSE remains above baseline at 10% and 20% data fractions, then drops abruptly below baseline at exactly 40% — 187 batteries for RWTH (+6.6%) and 16 batteries for XJTU (+2.6%) — with negligible further improvement at 60%, 80%, or 100%. This step-function profile is structurally distinct from the continuous improvement curves reported by Ren et al. [5]. HUST and MATR fail to beat baseline at any fraction, for reasons discussed in Findings 4 and 2 respectively.



(a) RWTH



(b) MATR

Fig. 3. RMSE heatmaps (window size compared to forecast horizon) for RWTH and MATR baseline sweeps. RWTH shows uniform low error across configurations; MATR requires window size of 50 cycles for reliable prediction.

TABLE III
BEST TRANSFER LEARNING RMSE IMPROVEMENT OVER BASELINE (%)

Target Source	CALCE	HUST	RWTH	MATR
XJTU	+20.6	+6.2	+1.2	−218
HUST	+21.2	—	+0.7	−240
RWTH	+21.7	−8.0	—	−253
CALCE	+5.6	−17.9	+6.6	−336
MATR	+21.3	−8.2	+6.9	—

Positive values indicate improvement; negative values indicate degradation. — indicates the same source and target dataset or a setting not evaluated.

D. Computational Cost Compared to Accuracy

Table VI reports training time alongside RMSE for all strategies on RWTH (XJTU source) — the largest target dataset and the one where cost differences are most pronounced. Head-only fine-tuning at 40% achieves near-baseline accuracy (RMSE=0.0900) in 525 s. Full fine-tuning at 40% is the only configuration to beat baseline (+1.2%, RMSE=0.0773) but at 1007 s — nearly double the cost. Full fine-tuning at 100% costs 3089 s while performing worse than full fine-tuning at 40%, confirming that additional data beyond the activation threshold provides no accuracy benefit

TABLE IV
CALCE TARGET: RMSE BY SOURCE, STRATEGY, AND DATA FRACTION

Source	Strategy	10%	40%	100%
XJTU	Head-only	0.2487	0.1692	0.1619
	Full fine-tune	0.2419	0.1706	0.1587
HUST	Head-only	0.2417	0.1697	0.1582
	Full fine-tune	0.2547	0.1747	0.1574
RWTH	Head-only	0.3969	0.1741	0.1620
	Full fine-tune	0.6974	0.1621	0.1565
MATR	Head-only	0.2392	0.1630	0.1589
	Full fine-tune	0.2394	0.1630	0.1573
Baseline (from scratch)		0.1998		
Zero-shot (best source)		0.3015		

TABLE V
CALCE SOURCE: FULL FINE-TUNE RMSE COMPARED TO DATA FRACTION

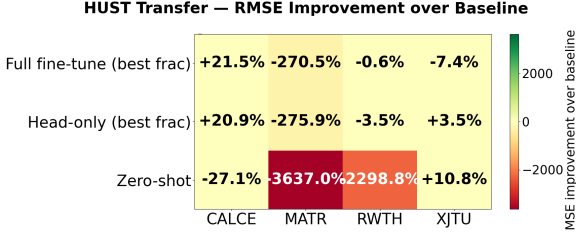
Target	10%	20%	40%	60%	80%	100%
RWTH	0.0865	0.0859	0.0730	0.0746	0.0761	0.0742
XJTU	0.2833	0.2632	0.2496	0.2666	0.2631	0.2629

Baseline: RWTH=0.0782, XJTU=0.2563. Bold: best per target. HUST and MATR fail to beat baseline at all fractions.

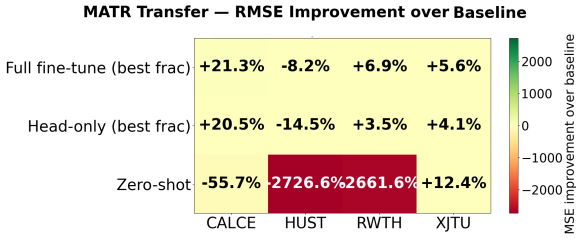
TABLE VI
RWTH TARGET (XJTU SOURCE): RMSE AND TRAINING TIME BY STRATEGY AND FRACTION

Strategy	Frac.	RMSE	Time (s)	compared to baseline
Zero-shot	—	1.8738	≈0	−2296%
Head-only	10%	0.0965	483	−23.4%
	40%	0.0900	525	−15.1%
	100%	0.0821	848	−5.0%
Full fine-tune	10%	0.0931	420	−19.1%
	40%	0.0773	1007	+1.2%
	100%	0.0833	3089	−6.6%
Baseline (from scratch)		396		0.0%

Bold: only configuration beating baseline. Baseline RMSE=0.0782.



(a) HUST



(b) MATR

Fig. 4. Transfer improvement heatmaps for HUST and MATR as sources. Both replicate the XJTU pattern: CALCE always improves (≈+21%), MATR always resists, confirming these are properties of the target dataset rather than the source.

at substantially higher computational cost.

V. DISCUSSION

A. Zero-Shot Deployment is a Reliability Risk

All 20 zero-shot evaluations produce catastrophic RMSE degradation. In V2G/V2B scenarios, a zero-shot model producing near-constant SOH estimates would fail to detect battery degradation entirely, creating direct safety risks for building electrical systems. The severity of failure correlates with domain shift — RWTH and MATR are the most resistant targets because their degradation dynamics are qualitatively unlike any source dataset — confirming that chemistry and protocol diversity compound the reliability risk.

B. Two-Regime Structure of Transfer Effectiveness

Transfer targets divide into two distinct regimes. *Source-independent targets* — CALCE and MATR — show improvement values essentially fixed regardless of source. CALCE achieves approximately +20% from every source (range: +20.6% to +21.7%); MATR resists transfer from every source

(no source achieves better than −218%). In both cases the outcome is determined by intrinsic target properties — data scarcity for CALCE, electrochemical uniqueness for MATR.

Source-dependent targets — HUST and RWTH — vary substantially with source choice. HUST ranges from −17.9% (CALCE) to +6.2% (XJTU), a swing of 24 percentage points; RWTH ranges from +0.7% (HUST) to +6.9% (MATR). For these targets, source selection is a meaningful engineering decision: the right source delivers reliable improvement, while the wrong source actively degrades performance. Notably, MATR and HUST share LFP chemistry yet MATR consistently resists transfer from HUST, confirming that chemistry alone does not determine transfer outcome — MATR’s fast-charge protocol encodes degradation dynamics absent from HUST’s multi-stage discharge cycling. This two-regime framing immediately tells a practitioner whether source selection matters for their target — something prior work evaluating only 2–4 pairs cannot provide.

C. Transfer Activates as a Step Function

The finer-grained CALCE sweep (Table V and Fig. 2) reveals that transfer effectiveness does not improve smoothly with additional data. For RWTH, RMSE remains above the from-scratch baseline at 10% (0.0865) and 20% (0.0859), then drops abruptly below baseline at 40% (0.0730, +6.6% improvement), with negligible further change at 60%, 80%, and 100%. The threshold crossing is clearly visible in the upper panel of Fig. 2 — the step-function profile is structurally distinct from the continuous improvement curves reported by Ren et al. [5], where transfer benefit accumulates gradually with additional target data.

For XJTU, shown in the lower panel of Fig. 2, the same threshold activation is observed at 40% (0.2496, +2.6% im-

provement). However, RMSE returns marginally above baseline at 60%, 80%, and 100% fractions. This behaviour arises because CALCE's three smooth LCO features cannot capture XJTU's high cycle-to-cycle variability — additional CALCE batteries reinforce representations that remain misaligned with XJTU's noisy NMC degradation dynamics. This confirms that reaching the activation threshold is necessary but not sufficient: the quality of the source feature space ultimately constrains the ceiling of transferable benefit.

Practitioners should therefore concentrate data collection on reaching the activation threshold rather than pursuing larger datasets beyond it.

D. Horizon Mismatch is a General Failure Mode

Prediction horizon alignment between source pre-training and target deployment is an independent reliability constraint — and a *general* failure mode rather than a dataset-specific observation.

Initial evidence came from RWTH as source (pre-trained at $H=25$): applied to $H=10$ targets, RWTH produced uniquely severe zero-shot failures — RMSE=1.712 on HUST, 1.819 on CALCE, 2.049 on MATR — substantially worse than XJTU and HUST sources at $H=10$ despite comparable feature overlap.

Independent confirmation comes from CALCE as source (also pre-trained at $H=25$). Applied to HUST ($H=10$), CALCE produces zero-shot RMSE=0.1279 (−1240% degradation). Critically, fine-tuning does not resolve this: at no data fraction does CALCE beat the HUST baseline, with best result RMSE=0.0113 at 100% data (−17.9%), versus +6.2% achieved by XJTU at $H=10$. Fine-tuning cannot undo the representational incompatibility introduced by horizon mismatch during pre-training. With two independent sources — RWTH (mixed chemistry, 668 cells) and CALCE (LCO, 6 cells) — exhibiting the same failure mode, horizon alignment must be treated as a first-order constraint on source model selection, independent of feature overlap or chemistry.

E. Feature Content and Strategy

MATR shares 44% of features with RWTH and XJTU yet resists transfer from all sources, because its 10 unique signals encode fast-charge LFP degradation modes absent from all source training sets. Feature content — whether shared features encode the target's dominant degradation mechanisms — is a more meaningful predictor than feature count. Table VI quantifies the strategy trade-off directly: head-only fine-tuning at 40% delivers near-optimal accuracy at roughly half the training time of full fine-tuning, and avoids the catastrophic forgetting risk that causes full fine-tuning to underperform head-only in low-data regimes.

VI. CONCLUSION

This paper investigated the failure modes and reliability boundaries of cross-dataset LSTM-based battery SOH transfer learning across 20 source-target pairs and 140 experiments.

Transfer targets divide into source-independent and source-dependent regimes, transfer benefit activates as a step function at a discrete battery count threshold, and prediction horizon mismatch is a general failure mode that fine-tuning cannot overcome. Zero-shot deployment is unsafe across all chemistry combinations tested. Future work will conduct multi-seed uncertainty quantification, investigate dataset similarity metrics as reliability predictors, and extend experiments to additional BatteryLife datasets.

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